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Concept of operations for increasingly autonomous space operations

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ABSTRACT

Concept of Operations (ConOps) documents provide a common view of future system functions to all stakeholders. This ConOps focuses on deep space missions, such as a mission to Mars. While Earth experts will be continuously monitoring operations during crewed deep-space missions, there will be communication delays and disruptions that will impede the rapid assistance required by the crew in time- and safety-critical situations. An argument will be made that the crew will require (some kind of) assistance to quickly understand the situation enough to safe the system. This document describes a notional vision of the operational processes, practices and capabilities needed by a deep space mission crew for them to autonomously respond to anticipated and unanticipated, time-critical anomalies. A descriptive model of a Crew Performance Support System (CPSS) is used to illustrate what will be required for a safe and successful manned mission to Mars. Scenarios will address crew, Earth-Support and technology roles/responsibilities, task prioritization, teaming strategies, complex procedure development and execution, assumptions, asynchronous collaboration under communication time delay and limited data exchange to illustrate potential operational needs and approaches. Scenarios are responsive to known human risks identified during and after long duration spaceflight and incorporate transition plans as space travel moves from ISS to Lunar to Mars operations specifically identifying test bed and research activity needs. The envisioned CPSS will alter the current operational paradigm of crew reliance on Earth experts to resolve anomalies. The intent of this ConOps is to advance the research and development of a CPSS.

1. Introduction

Today's crewed space missions are continuously monitored and provided with problem solving support by many Earth-based experts, experienced in a range of disciplines (engineering, science, medicine, etc.). This continuous support is possible because of the relatively close distance of spaceflights to date. Habitation outposts, such as ISS, Skylab, Mir, and Tiangong have orbited at 220 to 280 miles above the Earth. ISS operations, for example, rely heavily on the Mission Control Center (MCC) [1], Mission Evaluation Room (MER) [2], and Payload Operations and Integration Center (POIC) [3] expertise to diagnose and rapidly resolve anomalies. When an anomaly, or unexpected event, occurs, the problem is identified by MCC and then localized and resolved by the MER engineers (located in a room adjacent to the MCC). When an anomaly occurs during science execution, the MCC calls upon the POIC (located at Marshall Space Flight Center) to resolve the issue. The MCC then communicates a solution to the crew. The MCC, MER, and POIC will collectively be referred to as Earth-Support in this document. Continuous Earth-Support has been crucial for mission success and safety since unanticipated, critical malfunctions are typical occurrences. During the first six years of ISS habitation, for example, there were an average of 3 to 4 time-critical anomalies per year and critical malfunction alarms still occur several times a month [4].

As missions progress into deep space (i.e., regions of space beyond our Moon), communication delays and disruptions will restrict the crew's reliance on Earth experts. The distance between Earth and Mars, for example, can result in one-way communication delays of up to ∼20 minutes. Communication disruptions (e.g., communication equipment may be offline or temporarily nonfunctional) can also be expected. It is prudent to assume that critical, complex vehicle or habitat sub-systems will malfunction at a time when a deep space crew cannot rely on the Earth-Support team to detect, diagnose, and resolve the problem, and it is impractical to expect a small crew to step-in with the same level of expertise as 50 to 100 specialists on the ground. Pre-flight or in-flight crew training will not suffice for novel, unanticipated and time-critical risks with unknown ways to mitigate them. Earth return in case of an emergency will become less and less feasible

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the further the journey, and there will be limited resources from which to solve anomalies. The crew will need novel processes and capabilities to independently identify and resolve safety- and time-critical anomalies. At best, the crew will not be able to rely on Earth-Support guidance in the first ~40 minutes of time-critical anomalies. This demands that the crew act autonomously from Earth to resolve unanticipated, time-critical malfunctions and self-reliantly detect risk during safety-critical processes.

That a self-reliant crew may be unable to respond appropriately to time-critical anomalies has been identified as a significant risk to crew safety and mission success [5]. This risk is driven by several factors; novel and unanticipated anomalies would not have been trained pre-flight, the crew could forget their pre-flight training, spaceflight stressors could impair the crew's problem-solving ability, and current technological capabilities lack the maturity to aid the crew in unanticipated anomaly resolution [6]. Crew will need onboard, or habitat, capabilities to successfully respond to rapidly evolving anomalous situations. To reduce this risk will require not only a cultural change within space agencies, but also a concerted effort to raise the maturity and intelligence of crew information and decision support systems.

2. Approach

The approach taken to develop this Concept of Operations was multi-pronged:

- A Working Group was assembled and convened bi-weekly for 8 months. This group included experts in human-systems integration, human spaceflight, ISS anomalies, mission control operations, and crew training.
- Informal interviews of additional experts were conducted, including an astronaut, NASA Human Research Program researchers, NASA Element Managers, an NESC Technical Fellow, and intelligent systems software engineers.
- Published literature was reviewed including papers discussing the current maturity of intelligent systems, human anomaly response processes, NASA standards and requirements, and other NASA Concept of Operations.

The remainder of this paper will introduce a conceptual system that could support an autonomous crew traveling into deep space.

3. Crew Performance Support System (CPSS) description

3.1. CPSS Goals

There are ten ConOps strategic goals that focus on crew and Earth-Support personnel performance solutions for deep space operations.

1. **Improve Safety/Reduce Risk.** The CPSS should lessen the likelihood and/or severity of crew injury, crew loss, vehicle loss, and loss of mission due to the crew's inability to respond independently and autonomously as a team to safety- and time-critical anomalies.
2. **Enable Crew Autonomy.** The crew should be capable of, and empowered to, act independently of Earth-Support direction during space-to-ground communication delays and interruptions, routine tasks, and for time-sensitive operations. This new paradigm requires a robust human-systems integration approach.
3. **Incorporate a User-Centric Approach.** Human-centered design is an iterative design process where the design team (psychologists, domain experts, software and hardware engineers, crew, flight controllers and stakeholders) use their respective expertise to identify and address the user's needs [7]. The development of core components of the CPSS should employ human-systems integration design principles (this includes human factors) to balance cognitive and physical workload, situation awareness and trust [8] in the system. The approach will lead to a system that can be efficiently learned and operated, will reduce errors and error consequences, assist in the timely completion of tasks, and support the crew's ability to achieve the mission objectives. The CPSS should account for the various crew individual and team states, data entry and presentation interfaces (including multi-modal), computing platforms, and user preferences.
4. **Provide System Resilience.** Resilience describes the adaptive capacity and extensibility of a system to respond to surprises (e.g., unknown-knowns and unknown-unknowns) [9]. A resilient system provides assets with a broad set of uses to compensate for the constraints on resource availability and the mission's needs for asset functionality. The CPSS should strive to accommodate mission needs that fall outside its intended design. For example, the crew can think strategically and creatively, but they will be limited by what they can learn and remember during a time-critical anomaly, particularly when their performance has been compromised by spaceflight stressors. The crew should be able to rapidly request more, or less, support [10] and the CPSS should intelligently adjust the level of support [11] depending upon the user skill level and whether the crew may be physically, physiologically, cognitively, or behaviorally challenged. Data should be easily converted to information and that information should be easily re-organized to aid crew problem-solving.
5. **Provide Crew Training.** The CPSS should support crew knowledge acquisition and retention using in-habitat resources. When the crew is confronted with an anomaly, the CPSS should enable on-demand learning and recurrent training on:
   - Troubleshooting and problem-solving,
   - Spaceflight systems installation, activation, operation, inspection, maintenance, and repair,
   - Vehicle control, docking and habitat management,
   - Scientific experiment preparation, design, and implementation,
   - Medical and behavioral diagnoses and interventions (including medical task support),
   - How to effectively use the CPSS, such as procedure development aids, scheduling tools, inventory management or decision support tools, and
   - Teamwork amongst crewmembers, Earth-Support and the CPSS.

   The CPSS should provide an objective capability to track and provide feedback about crew proficiency and obtain training retention profiles. In the case of an anomaly, or anomalies, the system should provide support for the development of just-in-time training. This includes new knowledge and skills needed to resolve critical anomalies.

6. **Collaborate with the Crew.** The CPSS should adapt the information provided, based on situational data (e.g., sensor data, current crew tasks, inventory status, crew medical status, etc.) to produce flexible support and manage scarce resources. The maturity of today's intelligent systems and the trajectory of intelligent systems development suggests that total awareness of the mission context will not be possible. Given these limits, the system should accept contextual data directly from the crew. Efficient interactions and productive exchanges between the human and CPSS include timely information about the spacecraft, habitat, crew, and robots. The system should assist the crew in understanding the mission architecture and in-

teractions, permitting system state queries that identify potential causes for symptoms. The CPSS shares a special collaboration with the mission Commander, pushing pre-defined, mission critical status, including crewmember health and wellness. The CPSS and the crew should work in collaboration with a shared understanding of the task goals and subgoals, relevant contextual features, and existing scientific knowledge. The system could, for example, provide triage support (i.e., assign levels of priority to tasks or individuals to determine the most effective order in which to deal with them) when multiple problems have arisen. It should also designate an urgency status to alerts so that the crew understand which situation requires an immediate response. Just as human team members watch each other to coordinate their actions, the CPSS should continuously collect crew performance data so that it can perform as a team member. Using this performance data, the system could then adapt problem-solving and decision support accordingly.

7. **Collaborate with Earth–Support.** The CPSS should work collaboratively with Earth–Support. This includes maintaining Earth-Support situational awareness of flight activities and system status as communication constraints permit. This awareness permits Earth-Support teams to assess impacts of flight activities on mission goals and objectives and to be aware of when their support of crew is required. The CPSS should accept Earth-Support input or updates and alert the crew that new information has been provided.
8. **Provide Integrated Connections.** The CPSS should be an integrated component of the overall mission architecture. Integration should occur across all operational aspects of the system, including hardware, software, and human interaction. When allowed and appropriate, data collected by all domains should be sharable for collective use. To reach this goal will require a substantial advance in on-board data connections.
9. **Provide Intelligent System Support.** Outfitting the crew habitat(s) with technology, such as decision aids, is an attractive solution, however existing technologies are unable to effectively support crew anomaly response to unanticipated events [6]. Current technology can excel at tasks defined by rigid rules, but it falls far short of adapting to constantly changing conditions or operating in unfamiliar situations. The left side of Fig 1 (to the left of the solid red line) shows current-day progress in less-complex intelligent system development [12]. Most systems are reactive with limited memory, sometimes referred to as having "narrow intelligence".
   Spaceflight systems are very complex. Non-integrated ground-based testing of systems may not provide representative data for an intelligent system to learn. To progress will require the development of a mission data ingestion capability so that future spaceflight systems can learn about the influence of one system on another within the different, novel environments. Research needs to identify how much information the crew needs, what information is needed, how this information should be presented, and when to present the information. A far-reaching goal is to develop a system of systems that can learn and generalize knowledge. This scope is optimistic. Building system intelligence is more difficult than first imagined. The black, dashed line in Fig 1 shows a realistic improvement goal in system intelligence for a late 2030s crewed, deep space mission. However, this will only be possible with a significant and enduring boost in funding toward intelligent system development.
10. **Provide a Future Capabilities Test Bed.** The CPSS should serve as a platform for development and utilization of capabilities required to mitigate risks during manned deep space transport and surface missions. The in-mission system should be designed and built in a way that supports the continued development and use of components that expand autonomous capabilities and crew performance support. While the CPSS should be primarily developed to meet operational needs at the time, it should be built upon an infrastructure that supports an evolution of capabilities as they are needed for the current mission and as a test bed for capabilities needed for future missions. This is further discussed in the next section.

3.2. Critical Tasks

This ConOps is intended to describe crew performance requirements in time- and safety critical situations, when communication windows, latencies, disruptions, and potential failures will compromise crew access to, and availability of, Earth-Support personnel oversight. Fifty-five tasks that are likely to be especially challenging to a self-reliant crew were identified by an expert focus group consisting of crewmembers, flight/mission control, mission planning, medical, behavioral health, and training [13]. Of these, 11 were considered time-critical and 15 to be complex tasks. Table 1 presents three main goals and high-level task categories. These task categories were used for Section 4 scenario selection and to define CPSS functionalities.

To identify the crew's operational needs (e.g., processes, procedures, and technology) requires an understanding of how people problem-solve in time-critical situations. Investigators have observed workers in safety-critical occupations as they perform their jobs, including firefighters, NASA flight controllers in the MCC, astronauts on ISS EVAs, nuclear power controllers, medical personnel, and pilots [14]. The following section will briefly summarize human anomaly response based on this field research in the context of a self-reliant crew.

3.3. Human Anomaly Response

When an anomaly occurs, the status of an underlying state or process has changed (this could be with hardware, software, or a human physiological/behavioral process). To resolve the anomaly, the crew must first **detect** that an anomaly has occurred, or is occurring. This involves **recognizing** anomalous data patterns. Without real-time communication between the crew and Earth-Support, the crew will need to assess the immediate impact, safe the vehicle (or crewmember if it is a medical anomaly) and then determine a workaround. This is referred to as Failure, Impact, Workaround by the MCC [15].

The crew will then need to "track down" the anomaly source. **Troubleshooting**, or anomaly **diagnosis**, is a form of problem-solving that is logical and involves a search for the source of the problem. Initial anomaly diagnosis or classification (feature matching/story building) depends on expertise. Pre-flight and in-flight refresher training and experience will drive the troubleshooting process including what additional data should be sought to clarify the situation. Expectations provide a way to determine what changes (or the absence of changes) are informative and worth the crew's attention. Experts will typically choose an action that has worked in a similar situation. Crew knowledge, or limits to their knowledge, can frame how incoming information is interpreted.

Anomaly response is fraught with inherent uncertainties and trade-offs. When early cues strongly suggest plausible, but incorrect, diagnoses, later cues may be ignored or not given adequate weight. This is referred to as a Garden Path problem, where the initial problem-solving context may frame the crews' mindset in some direction that is inappropriate given the actual episode evolution. The operational context encourages fixation in that once the symptoms consistent with one diagnosis are observed, crew attention and activities may divert resources from situation assessment

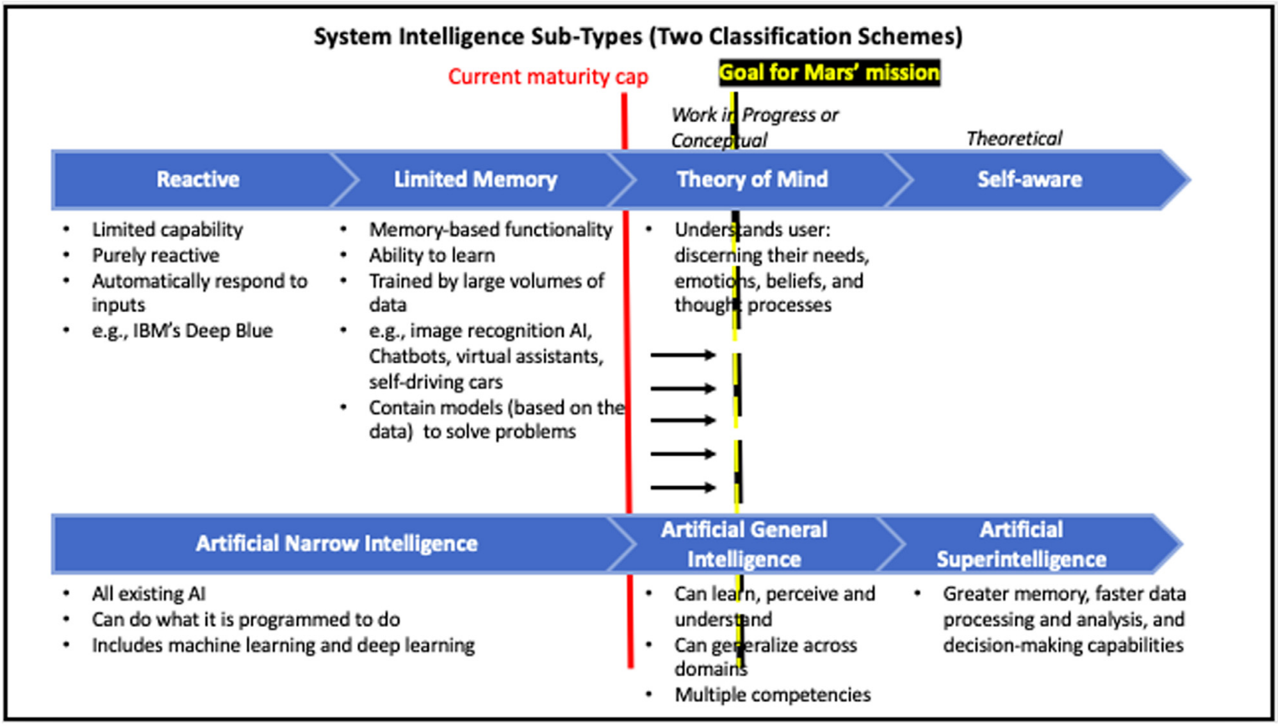

Fig 1. Two classification ontologies for intelligent systems showing the current maturity cap of these systems as well as what is needed to support a self-reliant crew on a deep space mission.

Table 1
Three exploration goals that include time-critical and/or complex tasks [13]. EVA = Extravehicular Activity or space walk.

Maintain vehicle/habitat	GOALS Perform mission-related tasks	Maintain crew health
Time-Critical Task Categories		
Respond to unanticipated major vehicle/habitat malfunctions	Perform repair and piloting/navigation task	Respond to medical/behavioral health events
Complex Task Categories		
Perform installations/activation/inspection	Perform EVAs	
Manufacture hardware, software, and fuel	Perform rover ops	

and additional hypothesis generation and evaluation. The diagnostic process used by the Mission Management Team (MMT) before the Columbia disaster demonstrates this phenomena (among other things), where they discounted uncertain evidence of trouble rather than vigorously following up [16]. Fixation can lead to missing, discounting, or reasoning away discrepant cues. Although experts typically take a suspicious stance toward any data that fails to fit the current assessment, there is a trade-off between the need to revise assessments and the need to maintain coherence. Vagabonding refers to the vulnerability of seeing every change as important or meaningful.

The initial interpretation and **response** effects how the anomaly progresses. Less appropriate, or timely, actions may sharpen difficulties, push the tempo in the future, or create new challenges. The level of crew expertise and knowledge – which could be enhanced by an information system - will determine their decision-making process. **Contingency management** consists of an analysis of possible damage or after-effects that may occur because of the anomaly and the crew's response to the anomaly. Anomaly **recovery** strategies are used to restore the fault to an acceptable state. The anomaly response processes of detection/recognition, troubleshooting/diagnosis, decision-making, response, contingency management, and anomaly recovery are not distinct, sequential stages, but rather they are interwoven processes that require revised situation assessments.

Processes, procedures, and technology must be developed to support the crew by augmenting each of these problem-solving processes. Optimally, a CPSS should act as an efficient, diligent, and effective team player. It should aid in early recognition of anomalous patterns without misdirection, rapidly provide safing procedures, help to diagnose the problem by provision of information, aid decision-making [17], work with the crew to predict how a response will affect the spacecraft systems, and provide long-term prognostic and recovery strategies. Because such a system is beyond the capabilities expected of intelligent systems, the following section describes a more conservative system, that with an appropriately focused research roadmap and sufficient funding, should be possible for deep space exploration plans.

3.4. A System that Supports Crew Anomaly Response in Deep Space Operations

3.4.1. Context

Fig 2 is a representation of the envisioned CPSS to support crew anomaly response. Anomaly resolution during a deep space mission will take place within a dynamic, or changing, context. In the upper left, Fig 2 shows five broad contextual variables: the organizational culture, mission context, system characteristics, team char-

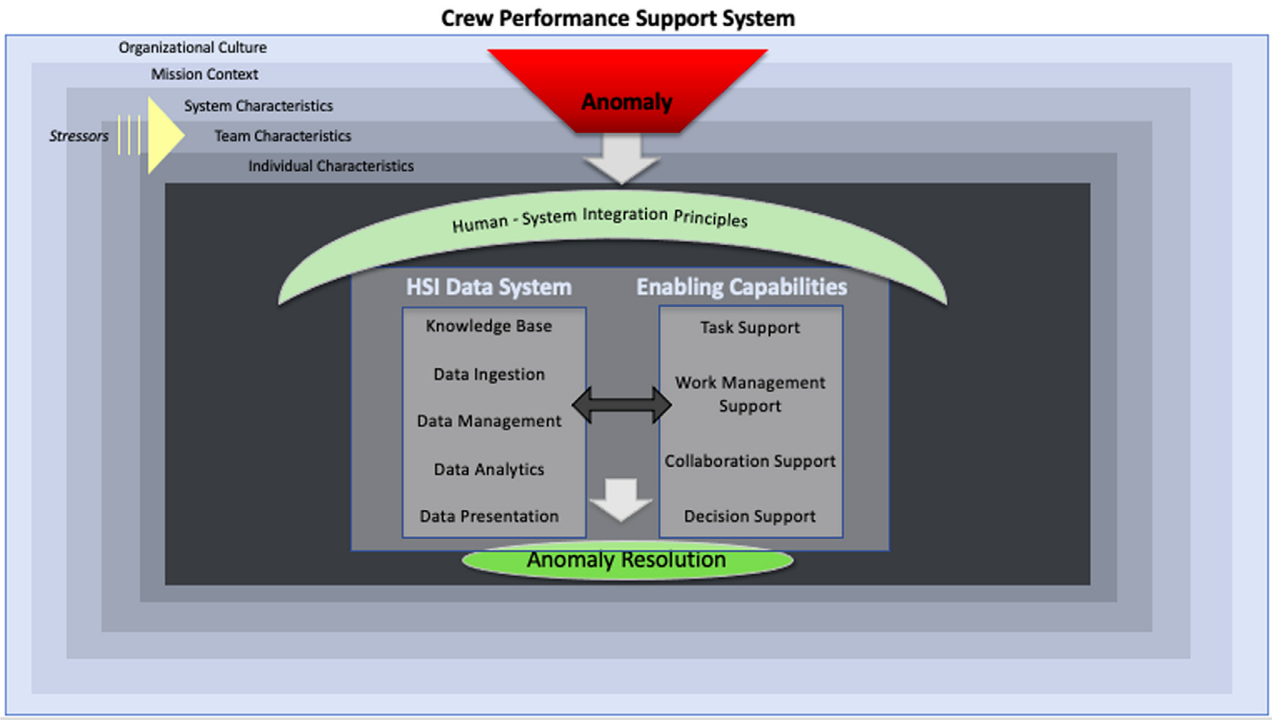

Fig 2. Theoretical characterization of factors that affect the crew's ability to self-reliantly respond to anomalies and how the Crew Performance Support System (composed of basic human-system integration principles, an HSI Data System and Enabling Technologies) can assist the crew in anomaly response. HSI = Human-Systems Integration

acteristics and individual characteristics. These contextual variables will now be described.

3.4.1.1. Organizational Culture. A space agency's or space industry's organizational culture forms the foundation of mission design decisions. The crew and Earth-Support are embedded in an organization which defines the budget prioritization, deadlines, and science return of a mission. Their high-level backing of a system to support crew anomaly response will govern what is made available to the crew. An example of the importance of considering organizational culture can be seen in causes attributed to the Genesis Probe Mishap during the "Faster, Better, Cheaper" era. The orientation and culture will directly influence the crew's goals, performance, and safety.

3.4.1.2. Mission Context. Mission context refers to, not just the mission phase (e.g., out-bound transit, surface operations, in-bound transit, etc.), but also the task(s) being performed. The immediate and upcoming situation will affect the crew's anomaly response reactions.

3.4.1.3. System Characteristics. System characteristics refer to how the crew habitat systems are designed and implemented. Examples include the presence, amount-of, and kind-of robotic support available to the crew to deal with different exigencies, types of accessible crewed rovers (e.g., pressurized, unpressurized) and the Crew Performance Support System maturity at the time of flight.

3.4.1.4. Team Characteristics. Team characteristics will also contribute to the crew's ability to appropriately respond to an anomaly. The crew, CPSS, and Earth-Support must work closely toward common goals. Collaboration tools must enhance trust and transparency, encourage alternative ideas, and rapidly adjust to role changes [e.g., 18].

Team interactions include:

- Crewmember - Crewmember (individual crewmembers interacting with one another),
- Crewmember - Technology (individual crewmember interacting with technology),
- Crew - Technology (several crewmembers interacting with technology),
- Crewmember - Earth-Support (individual crewmember interacting with Earth-Support),
- Crew – Earth-Support – Technology (several crewmembers, Earth-Support and the technology interact)
- Crew – Earth-Support (several crewmembers interacting with Earth-Support), and
- Earth-Support - Technology (Earth-Support interacting with technology).

Early in the transit into deep space, Earth-Support will be able to assist crew problem-solving in real-time. As the journey progresses, communication delays will not only change the cadence of support from Earth-Support, but stress from isolation and confinement, physical de-conditioning, psychosocial and interpersonal dynamics will become increasing concerns in the context of anomaly response.

3.4.1.5. Individual Characteristics. Individual crew characteristics will also contribute to anomaly response. When they are recruited, almost all astronauts have at least a master's degree in a Science, Technology, Engineering and Math (STEM)-related field. Those chosen for a deep space mission will undergo considerable pre-flight training on system hardware and software and at least one crewmember will be a physician. In addition to formal and pre-flight training, which could determine the effectiveness and efficiency of anomaly response, other individual characteristics that can contribute to anomaly response include a crewmembers' role and responsibilities, their metacognitive ability, their individual response to stress, their world model of how a spacecraft system functions and how it shall be used.

Table 2
Example spaceflight and task-related stressors that may occur on a deep space mission.

Spaceflight Stressors	Task-Related Stressors
• Atmospheric Toxins • Auditory Overload • Back Pain due to Time on Back • Catastrophic cost of making an error • Circadian Disruption • Communication Delays • Confinement • Cultural Differences between Crew • Decompression • Elevated Carbon Dioxide (CO_2) [19] • Fluid Shifts • Head Injury • High Workload [20] • Isolation • Lunar Dust • Radiation • Sensorimotor disturbances due to gravity transitions • Side Effects from Medications • Threat • Vision Changes	• Poorly organized information • Ambiguity • Conflicting Information • Evolving Scenarios • High Information Load • High Workload • Multiple Information Sources • Novelty • Performance Pressure • Time Pressure • Vendor design inconsistencies • etc.

3.4.1.6. Stressors. Fig 2 indicates that spaceflight and task-related stressors can affect system, team, and individual crewmember characteristics, and therefore anomaly response potential. Table 2 provides a listing of major stressors that can affect crew performance. Because stressors can disrupt physical and cognitive abilities, they can affect the crew's ability to use relevant information during an activity (e.g., the capacity to compare incoming information with information stored in long term memory).

In summary, there are at least five contextual characteristics that can influence crew anomaly response. The organization's culture and mission context are overarching forces. During the mission, system, team, and individual characteristics – which are affected by spaceflight and task stressors - will bound crew anomaly response. A Crew Performance Support System (CPSS) is required to aid the self-reliant crew. Observations of complex flight simulations and interviews with "front" and "back" room experts [21] identified potential anomaly resolution aids including information systems, advanced training systems, problem-solving assistance, and an overall goal of design for autonomous operations. Here we expand on this guidance.

3.4.2. Basic Human-System Integration Principles

The central portion of Fig 2 depicts a protective umbrella of human-system integration (HSI) principles that should be instituted during CPSS development [22]. While accounting for the situational context and the effect of stressors on system, team, and individual characteristics, the CPSS should employ basic human-system design principles in crew habitat and procedural designs. These principles include a balanced workload, shared situation awareness, and building an appropriate level of trust in the automation. For example, trust in a system is built when the system is reliable, transparent, and does not misdirect the user.

3.4.3. Crew Support Capabilities

Fig 2 also depicts two interrelated and cooperative components, an HSI Data System and other Enabling Capabilities. These components are an integral part of the teams listed above, greatly aiding a self-reliant crew to resolve anomalies and for Earth-Support to maintain awareness of the crew's situation (albeit with a time delay). The development of an HSI Data System and Enabling Capabilities will require a focused, intentional, and integrated plan across domains, elements, projects, and programs. The following two sections will describe these interrelated and cooperative system components.

3.4.3.1. HSI Data System. The CPSS should be based on a robust information technologies foundation to push toward flexibility and extensibility and facilitate development of data systems to support real-time decisions and data sharing. The HSI Data System consists of five components: a knowledge base, capabilities for data ingestion, data management, data analytics and data presentation. These components of the HSI Data System will now be described.

3.4.3.1.1. Knowledge Base. Even with extensive pre-flight training, to expect the crew to possess expertise in every sub-system and their interactions is unreasonable. When dealing with a time-critical problem, the crew may not possess memories of related problems that have been solved before. To support crew anomaly detection, the CPSS should contain sub-system data including hardware schematics, maintenance/repair/payload procedures, and software assumptions and constraints. The CPSS should contain expert knowledge and experiences (e.g., MER engineers, flight surgeons, etc.) and data from existing databases (Items for Investigation, Lessons Learned, Anomaly Reports, etc.).

3.4.3.1.2. Real-Time Data Ingestion. A system knowledge acquisition capability should automate the process of acquiring domain expertise, database, and real-time (e.g., sensor) data. In-flight ingestion of real-time contextual information could better support crew workload and situation awareness. Real-time data includes environmental data (e.g., CO_2 levels), alerts/warnings/alarms, crew schedule, crew clinical data, crew performance data and phase of flight. A system monitoring capability will be needed to automate the process of obtaining these data. Research will be needed to trade-off the benefits of these data for prediction with the invasiveness of the surveillance.

3.4.3.1.3. Data Management. The sheer amount of data in the Knowledge Base will be large. However, more data or information provided to the crew does not automatically lead to better decisions. Data Management includes issues related to understanding, updating, validating, cleaning, securing, transforming, enriching, and archiving the data. Data Management capabilities support crew workload reduction and information accessibility, availability, content, and transparency.

3.4.3.1.4. Data Analytics. The envisioned system should overcome the limitations of an unaided crew, physiologically, biomechanically, perceptually, and intellectually. Data Analytics should support anomaly detection and recognition, thus relieving the crew from the need to closely monitor the systems. A data integration process should harvest (i.e., produce a unified view of) the data within the knowledge base and the real-time data (integrated sensor data) that has been ingested.

Data Analytics will support failure diagnostics. Anomaly diagnostics should contribute to information quality and interpretability through the extraction of patterns and trends in the data (e.g., pattern recognition, outlier detection, anomaly recognition). To diagnose a situation and predict what could occur in the future requires a system that can cope with complexity. For example, the system should be capable of developing an appropriate description of the situation (i.e., the context) and find ways to reach the current goals. Diagnostics also includes anomaly isolation and identification. For example, in the case of a hardware failure, full diagnostics requires determining the specific fault mode, rather than just which sensor has an unusual value. The CPSS should contain nominal and off-nominal thresholds for each sub-system.

Data Analytics will also support failure prognostics. Prognostics involves detecting the precursors of a failure and predicting how much time remains before a likely failure. The system should be

Table 3
Enabling Capabilities in the Crew Performance Support System.

Subdomain	Description
Task Support	Capabilities that link integrated system resources, conditions, and performance monitoring with the presentation of clear, well-structured, easy to consult information. Monitoring would be executed on the crew, hardware, software, and the environment.
Work Management Support	Capabilities that apply the crew's workflow to the movement of information. Work management transforms and streamlines crucial processes by integrating processes and procedures needed by the crew, such as integrating the activity schedule with computerized procedures and the current inventory availability and location.
Collaboration Support	Capabilities that enhance crew-crew, crew-Earth-Support and crew-automation interaction and working as a group. The technology facilitates the sharing and distribution of knowledge and expertise among the relevant community (e.g., EVA crew, Intravehicular Activity (IVA) crew, Earth-Support). Collaboration Support helps in the development of a shared understanding of a problem and its solution process. It also captures and models information and knowledge of group activity and uses it to achieve more effective group monitoring and support.
Knowledge-and model-based Decision Support	Capabilities designed to support more effective and efficient decision-making through timely and appropriate data, information, and knowledge management. In addition, these systems support decision-making through prediction, diagnosis, and recommendation/resolution techniques.

tuned to the future, providing the crew with what can happen next.

3.4.3.1.5. Data Presentation. The CPSS interface should amplify cognitive functions needed for anomaly response (e.g., detection, discrimination, interpretation). The crew should be able to notice what they are not expecting to see. The system, for example, could represent spacecraft systems, through displays (e.g., visual graphics). How the information is displayed defines the crew's understanding. The system must show a certain level of transparency to balance crew trust and task complexity. Often technological improvements are predicated on what technology can do rather than on what the humans need. The inevitable result is that the human becomes the system bottleneck. The automation, operations and work environment should match human capabilities and limitations. Thus, as the system is developed, simulations should reveal changing demands on the crew from the increasingly complex technological system.

The right-hand side of Fig 2 contains a box labelled *Enabling Capabilities*. In conjunction with the HSI Data System, additional enabling capabilities will be required to support crew anomaly response and Earth-Support situation awareness. These additional capabilities will now be discussed.

3.4.3.2. Enabling Capabilities. The Crew Performance Support System is composed of four enabling capabilities that capture potential anomaly resolution aids. These capabilities are summarized in Table 3 and further described below.

Fig 3 provides an alternative *Enabling Capabilities* representation. This figure shows that as we journey from ISS to the Moon to Mars that multiple, simultaneous, and integrated research and development efforts (i.e., support systems co-evolution) must be implemented to meet the problem-solving challenges a self-reliant crew will face on a deep space mission. The crossovers between the capabilities (shown with bidirectional arrows) are just as important as the discrete capabilities themselves. As the capabilities mature, the lines between the support subdomains will blur and an integrated system will emerge. The processes, practices and technological goals shown in italics in Fig 3 are representative and not intended as sole solutions.

3.4.3.2.1. Task Support (TS). The first enabling capability is referred to as task support. A task support (TS) system will aid anomaly detection, recognition, troubleshooting, response selection, and contingency management. This support could include such aids as system monitoring which refers to hardware or software components used to monitor system resources, conditions, and task performance. System monitoring is the backbone of anomaly detection. On an unprecedented mission to Mars, system hardware, software, and crew health and well-being could be continuously monitored. A far-reaching goal would be for monitoring techniques to automatically learn and tune themselves and potentially identify unknown failures, or those not previously identified and categorized. System monitoring will generate a tremendous amount of data that must be managed and stored in the Data System. Procedures, recurrent and just-in-time training, augmented (possibly multi-modal) displays, inventory tracking, and knowledge augmentation are forms of task support.

3.4.3.2.2. Work Management Support (WMS). WMS includes capabilities that apply the crew's workflow to the movement of information. Crucial processes are streamlined, such as integrating the activity schedule with computerized procedures and the current inventory availability and location. Work management support will aid individuals and teams to organize their workflows by bringing visibility to team members across all levels. Work management helps teams reach their goals (i.e., anomaly detection, troubleshooting, response selection, contingency management, and restoration) more efficiently.

3.4.3.2.3. Collaboration Support (CS). The processes and tools developed under this enabling capability facilitate real-time and asynchronous communication between for example the EVA crew, Intravehicular Activity (IVA) crew or Earth-Support. Collaboration Support helps in the development of a shared understanding of a problem and its solution process, supports consensus building, streamlines resource management, and enables remote presentation and archiving.

3.4.3.2.4. Knowledge- and Model-Based Decision Support (DS). This capability should be designed to support decision-making by effectively processing timely and appropriate data, information, and knowledge management. These types of systems may support decision-making through diagnosis, prediction, or recommendation techniques. For example, electronic systems exist (e.g., ADAPTS [23]) to support maintenance technicians that integrate adaptive guidance from diagnostic systems with adaptive access to technical information (i.e., supporting what to do and how to do it). This type of capability could require that crew models be represented in domain or task models.

3.4.4. How the CPSS Interfaces with other Systems

The CPSS will exist within a broader architecture. To aid crew in reacting to events and maintaining situation awareness, all exploration technologies (typically based on technical disciplines) must recognize that if that system fails, consequences go beyond its own confines. Coupling and dependency among technologies is significant for design, operations, management, and maintenance of the sub-systems. Table 4 lists these interfacing technological subsystems as well as how that system could augment CPSS function.

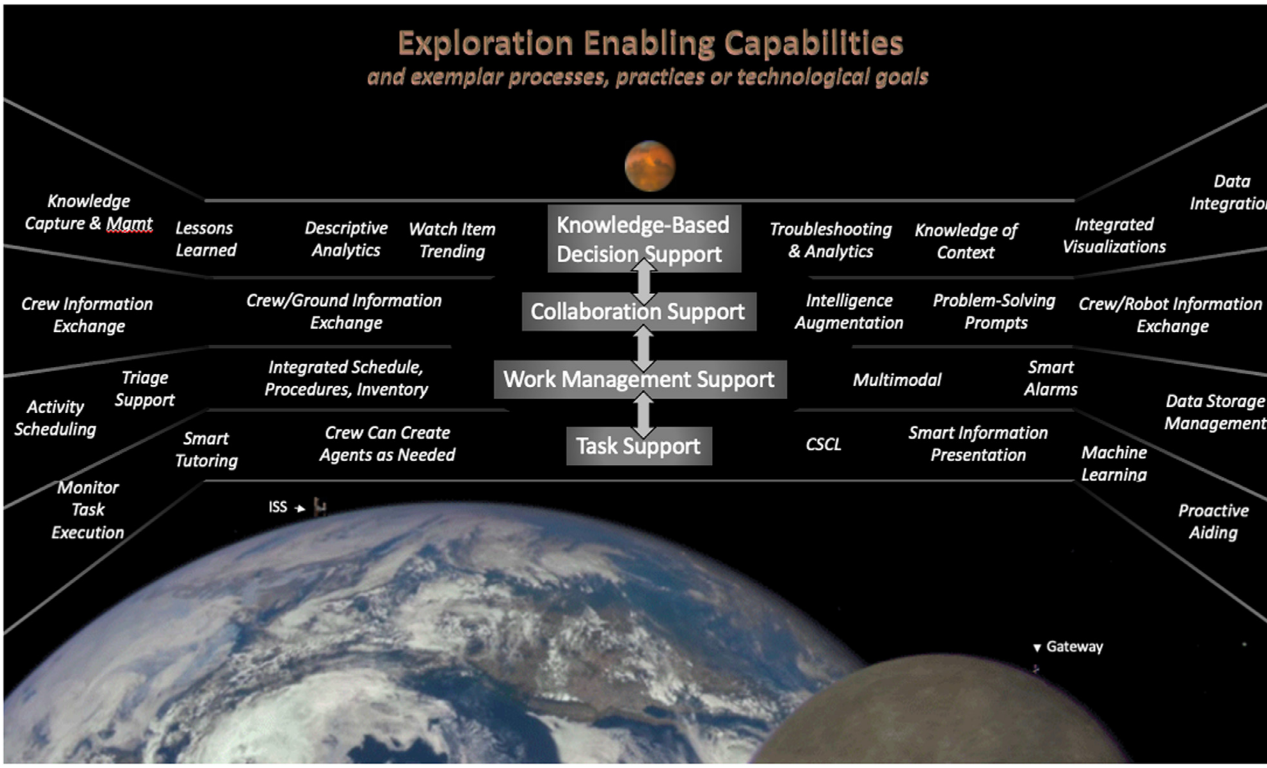

Fig 3. Conceptual depiction of technological needs for a deep space mission, such as a mission to Mars. The types of problem-solving support that will be needed by a self-reliant crew are divided into four enabling capabilities: Task Support, Work Management Support, Collaboration Support and Decision Support. Examples of each type of support are listed in italics. CSCL = Computer-assisted Collaborative Learning.

Table 4
List of exploration technologies and their interface with the CPSS.
Technology list is from [24]

Interfacing Technology	CPSS Interface Essentials Levied on other Systems
Propulsion	Provides CPSS data or information needed for crew situation awareness about space launch and in-space propulsion logistics, resources, and effects on crew (i.e., gravitational forces, vibration, chemical leaks, electrical discharge, etc.).
Flight Computing and Avionics	Provides processed data from smart sensors and instruments to the crew and provides telemetry to Earth-Support, including data on the status and health of vehicle systems, system trending data, and notifications of off-nominal and caution and warning events.
Power and Energy Storage	Provides situation awareness to crew about power generation, management, or distribution issues as well as current energy stored and projected to be used within a defined time. Accepts estimates of power needs for crew tasks to calculate projected availability.
Robotics	Provides sensing and perception data to crew as appropriate, safe mobility and manipulation practices, autonomous aiding for crew workload reduction while keeping crew in the loop. Provides safe and efficient human-robotic interaction.
Communications, Navigation & Orbital Debris Tracking	Provides optical and acoustic communication with other crew and with Earth-Support. Transfers crew commands, spacecraft telemetry, mission data. Informs crew of communication latencies and outages, and position, navigation, and timing situation awareness. Provides space debris situation awareness.
Human Health, Life Support and Habitation Systems	Provides life support and environmental monitoring, control, and situation awareness (i.e., interfaces) for all crew habitat systems. Examples includes radiation and elevated CO_2.
Exploration Destination Systems	Provides crew with technologies to enable successful crew task activities during the mission including in-situ resource utilization.
Sensors and Instruments	Provides the CPSS with sensor data within and outside of the habitat.
Entry, Descent & Landing (EDL)	Provides CPSS data or information needed for crew situation awareness about EDL logistics, resources, and possible effects on crew (i.e., gravitational forces, vibration, etc.).
Autonomous Systems	Provide crew with situation awareness of autonomous operations.
Software, Modeling, Simulation, and Information Processing	Provides software, modeling, and simulations for the advancement of intelligent systems to support crew.
Materials, Structures, Mechanical Systems and Manufacturing	Provides technologies and procedures to aid in the development and manufacturing of new materials on lunar and Mars surface.
Ground, Test and Surface Systems	Provides technologies and procedures for assembly of surface habitats.
Thermal Management Systems	Provides crew with situation awareness about thermal system status.
Range Tracking Systems	Provides situation awareness of weather (e.g., lunar dust storm activity).
Guidance, Navigation & Control	Provides guidance and targeting decision support.

3.4.5. CPSS Iterative Development

Incremental missions to the Moon should advance and validate capabilities required for humans to live and work in deep space. Current plans are for the first few lunar missions to focus on spacecraft demonstrations followed by increasingly longer crewed missions. An incremental buildup of infrastructure on the surface will follow for the next ten years including a Base Camp with rovers, power systems and ambulatory habitats. Robots will work with the crew to search for valuable resources, such as water – which can be converted to oxygen and fuel. During this time, landing technologies and advanced mobility capabilities will permit farther excursions to unexplored regions of the Moon. CPSS technologies needed for a journey into deep space should be demonstrated during these sustained lunar missions.

To prepare for lunar demonstrations, the next several decades should be used to design, develop, and iteratively test the needed capabilities for a self-reliant crew on deep space missions. There are currently no autonomous systems that can solve novel, complex problems. These systems will need to recognize patterns, read smart sensors, and analyze big data. To support crew anomaly response will require an intensive effort and unprecedented backing. Iterative development should start with walkthroughs, storyboarding and laboratory testing. Concepts can then be tested in space analogues (e.g., NEEMO) and the usefulness of information conveyed by the system verified by MER engineering experts. Just-in-time training can be validated within crew training facilities. Finally, mature concepts should be tested in high-fidelity environments such as on the ISS, Gateway, or other Lunar missions.

As an example of the essential, iterative intelligent technology development process, a knowledge capture capability must be instituted to secure past and ongoing operational and engineering experience. First, the database technology could be adapted to readily accept and store existing spaceflight knowledge, lessons learned, flight rules, payload, and service procedures, etc. This database could then be advanced to receive real-time data from critical systems and sub-systems, such as smart environmental sensors and non-invasive task performance data. The reliability and validity of those data must be tested in increasingly more complex problems in progressively higher fidelity simulations. In summary, individual capabilities should be initially developed and evaluated, followed by the co-evaluation of networked, or combined capabilities.

Each capability should co-evolve. For example, a data analytics capability could be developed to process and interpret the database records. Sub-components of an overall, integrated CPSS should then be evaluated in medium fidelity analogues and additional, essential capabilities identified. Finally, as the first manned Mars or other deep space mission draws closer, sustained Lunar missions offer an environment to gain critical experience and testing with the integrated CPSS to ensure crew can safely and autonomously perform unplanned, safety- and time-critical tasks. Lunar testing is mandatory since once the system has been used in operations the increased system complexity invariably leads to new and unexpected problems [25].

4. Scenarios

The following scenarios assume that there are four crewmembers (see Fig 4). On the journey to Mars, the Commander is Bill. He is a certified pilot and clinical psychologist. He has also undergone significant medical training as a Crew Medical Officer (CMO), although he is not a physician. Dimitri, who is also a certified pilot will be referred to as Flight Engineer 1 (FE-1) on the trip to Mars. He is the Commander on the return to Earth. He has master's degrees in mechanical and electrical engineering and has received extensive training on the maintenance and repair of the space systems. He is also astoundingly handy and creative. FE-2 is Helen. She has a doctorate in planetary geology, is an experienced researcher and has also undergone extensive medical training to step in as CMO as required. Christian, FE-3, is a certified pilot and emergency room physician and will serve as CMO for the mission. Christian also has experience with traumatic event counseling. They are a culturally diverse crew that have interacted affably and effectively during their five years of training for this mission. They can all speak fluent English, Russian and German.

The launch was delayed three times for various reasons and as a result the crew waited in the *Numa* crew vehicle for up to five hours on each abort attempt. Bill experienced lower back pain on the second and third abort attempts that he rated as a 7 on a ten-point pain scale.

The *Numa* habitat and the *Fasti* service module are nearing the halfway point on their cruise to Mars. All the crew miss their families and friends, but are coping well. Regular assessments of crew visual function have indicated some retinal changes in all crewmembers. FE-2 is showing a significant decrease in far and dynamic acuity.

4.1. En route to Mars

The *en route* transit to Mars will use the Deep Space Network for communication. This means that communication will be limited, e.g., no real-time communication can be assumed due to propagation delay as well as scheduling of communication assets

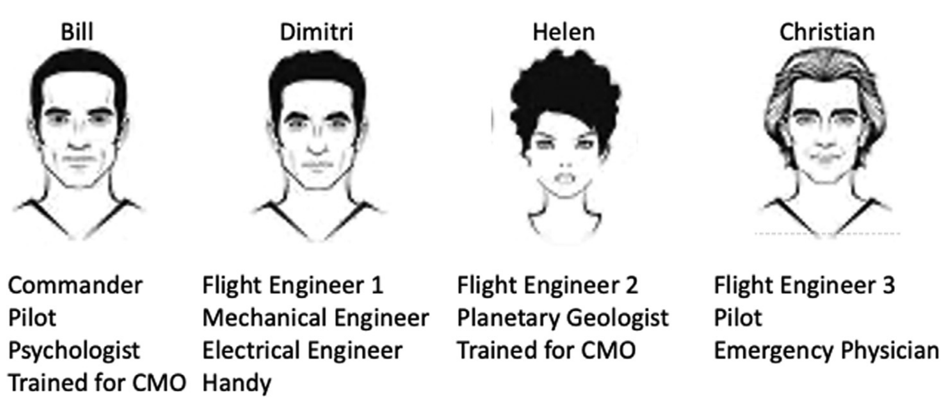

Fig 4. The four crewmembers in the Numa crew vehicle on their way to Mars

Table 5
Cometary Debris Field Event
Each row in the following table represents a point in time, with time flowing downward in the table. Anomaly A (first column) refers to the onset and resolution of a single anomaly. The column labelled Anomaly B refers to a second, concurrent anomaly. The third column identifies the CPSS enabling capability (see Section 3.4.3.2) and a representative technological function. TS = Task Support; WMS = Work Management Support; CS = Collaboration Support; DS = Decision Support. The rightmost column provides the exploration technology for which the CPSS has levied a requirement. These disciplines were described in Section 3.4.4, Table 4. Bold font is used for Anomaly A-relevant CPSS capabilities and exploration technologies.
The space capsule is *en route* to Mars and has flown through an undetected cometary debris field.

Anomaly A: Loss of Power – Scenario inspired by [27]	Anomaly B: Loss of Cooling – Scenario inspired by [13]	CPSS Enabling Capabilities and *Technology (italics)* demonstrated Bold text = Anomaly A	Exploration Technology for which the CPSS levies requirements
Smart sensors isolate the source of the issue, integrating a potential cascade of alarms into a single auditory alarm [28] accompanied by a message indicating that the habitat and service module are experiencing a significant power reduction.		**WMS (*Smart Alarms*)** **WMS (*Multimodal Alerting*)**	**Power & Energy Storage** **Sensors & Instruments**
The CPSS event recording capability is automatically triggered by the alarm onset e.g., records audio, keypresses, data, etc.		**TS (*Event Driven Recording*)** **TS (*Proactive Aiding*)**	
Power & Energy Storage and data analytics determine that life support power needs will not be met in ~5 hours at the current rate of loss. These data are pushed to the CPSS Data System. The CPSS presents these data and the trend graphically to the crew.		**TS (*Proactive Aiding*)** **DS (*Models/Descriptive Analytics*)** **DS (*Integrated Visualization*)**	**Power & Energy Storage**
FE-1 selects the Watch Item icon indicating that this trend must be monitored.		**DS (*Watch Item*)**	
The WMS capability lists 10 items using power that may be turned off without immediate consequence (e.g., won't compromise an experiment) to conserve power. FE-1, immediately turns OFF eight of these options, choosing to maintain full power to the CO_2 scrubbers for now. This change in state automatically triggers a recalculation of the power loss trend watch item.		**TS (*Proactive Aiding*)** **WMS (*Monitor Task Execution*)** **DS (*Knowledge of Context*)**	
Prior to the mission, previous spaceflight experiences, spaceflight data, lessons learned, procedures, etc. were acquired and stored. FE-1 performs a digital search for historical records relating to a loss of power by stating "Critical loss of power". Similar spaceflight experiences are automatically identified from the knowledge database. Selectable links to three historical records are presented to FE-1. FE-1 scans the 3 historical records and decides that the entry describing a 2007 event of a torn solar array during its deployment best corresponds to the current loss of power trend. FE-1 shares this historical record with the entire crew.		**DS (*Knowledge Capture & Management*) (*Lessons Learned*)** **TS (*Voice Recognition*)** **CS (*Crew-Automation Information Exchange*)** **DS (*Knowledge of Context*)** **DS collaborates with WMS** **WMS (*Data Management*)** **TS (*Proactive Aiding*)** **CS (*Problem-Solving Prompts*)** **CS (*Crew-CPSS Information Exchange*)** **CS (*Crew Information Exchange*)**	
While viewing the communication pane, FE-1 sees that Earth-Support will receive notice of the ensuing power loss and trending data in approximately 15 min		**CS (*Maintain* Earth-Support *Situation Awareness*) i.e., Provides information about when messages will be received on Earth**	**Communications**
FE-1 selects access to the exterior camera feeds. He first accesses the port-side camera but does not see an obvious problem. He switches the image to the starboard-side and detects a large dark spot on one of the solar arrays adjacent to the junction box/power harness. The starboard solar array has been damaged by debris. He shares this information verbally. Earth-Support is automatically sent the images and the audio recording.	Suddenly, a warning alarm and message alerts the crew that a loss of pressure in external cooling loop B has been detected.	**TS (*Smart Information Presentation*) Presents exterior camera views with rapid switching between cameras.** WMS (*Smart Alarms*): e.g., Single alarm (rather than a cascade of alarms) that alerts the crew to a significant loss of cooling	**Sensors & Instruments** **Communications**
Although momentarily distracted by the second alert, FE-1 continues to read through the historical record and learns that the 2007 crew creatively used a cuff-link type device to pull and secure their solar panel rip.	The Commander states "Helen and Christian, please address that alarm"		

(*continued on next page*)

Table 5 (*continued*)

Anomaly A: Loss of Power – Scenario inspired by [27]	Anomaly B: Loss of Cooling – Scenario inspired by [13]	CPSS Enabling Capabilities and *Technology (italics)* demonstrated Bold text = Anomaly A	Exploration Technology for which the CPSS levies requirements
FE-1 recalls a similarly shaped device onboard, calls up the inventory tracking system and locates the item. FE-1 sends a text message that identifies the item to the Commander.	FE-3 requests a diagnostic on the external cooling system	**DS (*Knowledge Capture*)** **WMS (*Inventory Tracking*)** **CS (*Crew Information Exchange*)**	Thermal Management System
	FE-2 visually confirms and verbally states “Ammonia crystals near the external cooling loop”.	CS (*Crew Information Exchange*)	
The Commander checks the loss of power trending graph watch item and sees that they now have 4.5 hours until life support power needs will not be met. He also sees that while Earth-Support is now aware of the power loss anomaly, they are not yet aware of the cooling loss.	FE-3, the physician, knows that FE-2 has been experiencing vision problems and provides a secondary confirmation of the ammonia crystals “Concur Helen, ammonia crystals”	**DS (*Watch Item Trending*)** **CS (*Crew-Earth-Support Information Exchange*)** CS (*Crew Information Exchange*)	**Communications**
The Commander requests a comparison between the two anomalies.	The Commander requests a comparison between the two anomalies.	**WMS (*Triage Support*)**	
The CPSS shows a prioritized listing of all current activities (including those suspended when the first alarm sounded) in order of criticality. Next to each activity is the rationale for its priority in the list. The CPSS has identified the power loss anomaly as the number one priority.	FE-2 works with the system to reroute critical systems to Cooling Loop A to safe the vehicle.	WMS (*Triage Support*) DS (*Knowledge Capture) e.g.,* Event records are linked to event solutions	Thermal Management System
	The Commander confirms with the FE-2 and FE-3 that, although the loss of cooling is a critical issue, it is not immediate.	CS (*Crew Information Exchange*)	
The Commander decides to prepare for an immediate EVA to repair the solar array tear. He verbalizes this decision to the crew.		**CS (*Crew Information Exchange*)**	
The Commander opens the text message from FE-1 and asks FE-2 to retrieve the cuff-link type device to pull and secure the solar panel tear.		**CS (*Crew Information Exchange*)**	
FE-3 views the procedure used by Scott Parazynski during the 2007 repair. He converts the procedure into an editable form and the system provides a prompt to rename the file. The Task Support procedure writing capability works with FE-3 and FE-1 to customize the procedure for the current situation and combine it with the procedure for junction box diagnosis and repair.		**DS (*Knowledge Management*)** **TS (*Proactive Aiding*) e.g., link to historical procedure** **WMS (*Procedure Writing Tool*)** **TS (*Word Processing Prompts*)**	
In the meantime, FE-1 and FE-2 work through the pre-EVA procedure.		**WMS (*Computerized Procedures* [29])**	
The Commander adds two new tasks (Criticality 1 - EVA-A to mend the solar array and Criticality – 2 - EVA-B to fix cooling loss anomaly) into the self-scheduling tool which inserts the tasks into the timeline.		**WMS (*Self-Scheduling Tool*)**	
The Commander and FE-1 successfully repair the solar array tear.			
Power & Energy Storage detects a normal power range and sends a message to the CPSS and to the Commander. The CPSS suggests discontinuing the automatic event recording capability. FE-2 concurs. The time-critical anomaly has been resolved.		**TS (*Event Driven Recording*)** **TS (*Proactive Aiding*)**	**Power & Energy Storage**
After doffing the EMU, the Commander reports grade 5 of 10 lower back pain to the CMO who enters this into the Commander’s medical record.		**DS (*Knowledge Management*)**	
	The following day, the crew turns to the work of repairing the leak and repressurizing Cooling Loop B.		
	Ground-Support has up-linked a computerized procedure and training for the leak repair.	CS (*Crew*-Earth-Support *Collaboration)*	Communications
	The FE-1 and FE-2 work with the system to edit the EVA training and execution plan. They downlink the edited plan to Earth-Support.	TS (*Computer Supported Collaborative Learning*)	Communications

(*continued on next page*)

Table 5 (*continued*)

Anomaly A: Loss of Power – Scenario inspired by [27]	Anomaly B: Loss of Cooling – Scenario inspired by [13]	CPSS Enabling Capabilities and *Technology (italics)* demonstrated Bold text = Anomaly A	Exploration Technology for which the CPSS levies requirements
	The next day, the Commander and FE-1 train for EVA-B using the onboard virtual reality training system with a robot avatar (controlled by FE-2) to run through the EVA and test the plan and procedures.	TS (*Computer Supported Collaborative Learning*)	Robotics
	As usual, Earth-Support has included a list of needed tools at the beginning of the procedure. FE-3 works the EVA Tool Gather activity. FE-3 requests the location of the needed tools, and the Inventory Tracking and Activity Scheduling capabilities provide her with a list of the locations for efficient gathering.	CS (*Crew*-Earth-Support *Collaboration*) WMS (*Inventory Tracking*) WMS (*Activity Scheduling*)	
	Based on their training runs, the Commander asks FE-2 (the crew member assigned to the robot) to command a counter hold for the robot for the clamp installation.	CS (*Crew Collaboration*)	Robotics
	The EVA-B crew members further refine the procedures and plan, adding an image to a procedure step and modifying the schedule to start EVA-B 30-minutes earlier than planned.	WMS (*Computerized Procedures*) WMS (*Self-Scheduling*)	
	FE-3, the CMO lead, conducts a physical on the Commander, FE-1, and FE-2. Although tired, he thinks they are all OK for tomorrow's EVA. The CMO provides them with sleep medication.	TS (*Medical Procedure*) WMS (*Medical Inventory Tracking*) DS (*Knowledge Capture*) e.g., inputs use of sleep medication into personal medical records	
	During pre-sleep the onboard lighting adapts to the change in the sleep schedule for the EVA.	WMS (*Scheduling Tool*) communicates with the Wellness System (*Sleep Management Support*)	
	Two days after the failure, the crew wakes up refreshed and ready for the EVA. The CMO prescribes mild pain medication to the Commander to prevent the back pain he reported after EVA-A.	WMS (*Medical Inventory Tracking*) DS (*Knowledge Capture*) e.g., inputs use of pain medication into personal medical records	
	FE-2 sets up her procedures and the live camera feeds at the robot station as the CMO prepares to monitor each EVA-B crew member's medical status.	WMS (*Computerized Procedures*) CS (*Crew-Robot Collaboration*)	Robotics
	During EVA-B, the IVA lead (i.e., the FE-3) confirms the EVA crew members and robot's steps. He calls out each vehicle command, confirming that each step is complete, until the repair is finished, and coolant is routed back into loop B.	CS (*Crew Information Exchange*)	Robotics
Each of the crewmembers involved in the anomaly input their lessons learned (LL) into a LL Database.	Each of the crewmembers involved in the anomaly input their lessons learned (LL) into a LL Database.		

After five very stressful days ending with two successful EVAs, Commander Bill, at the CMOs advise, gives the crew an off-duty day. Houston sends a congratulatory video to the crew from the flight control team. On their day off, the crew choose to read their personal correspondence in the morning and to watch a movie together after dinner.

[26]. Data files sent to Earth-Support will consist of video, voice messages, and telemetry. Spaceflight stressors, such as deep space radiation, isolation, microgravity, communication delays and no resupply are expected to affect crew anomaly response on the transit to Mars.

It is mid-morning, and all crewmembers are busy with their scheduled activities. Suddenly, they are passing through an undetected cometary debris field that triggers two separate anomalies. Table 5 shows the CPSS and crew responses for each anomaly in separate columns. The leftmost column describes a "Loss of Power" scenario. The second column describes a "Loss of Cooling" scenario which partially overlaps in time with the loss of power. Each row shows successive points in time. The third column lists the CPSS enabling capability:

- Task Support (TS),
- Work Management Support (WMS),
- Collaboration Support (CS), and/or
- Decision Support (DS).

In italics is a theoretical intelligent system that could assist in that task (from Fig 3). These specific technologies are representative and are not intended to indicate requirements. The exploration technology (from Column 1 of Table 4) for which the CPSS levies a requirement (e.g., requires data from) is provided in the rightmost column, as applicable.

It is suggested that the reader first examine Anomaly A: Loss of Power.

4.1.1. Event: Cometary Debris Field (Unanticipated)

The two scenarios shown in Table 5 demonstrate the complexity of operations and how a crew performance support system could assist a stressed and busy crew. With a crew size of four, workload is very high. There would likely be many more issues simultaneously occurring. For example, during the power loss scenario a reduction in overall power usage could be implemented by the system including reduced lighting, etc. If air circulation or CO_2 scrubbing has been reduced, then the system could alert the CMO when a threshold-level of CO_2 has been reached (e.g., possible cognitive challenges). In addition, the system could collate the streaming and verbal data to produce an incident report enabling the crew to resume other duties.

The next section assumes that the crew has successfully reached Mars and have descended to the surface.

4.2. Mars Surface

Once safely on the surface, crew will perform sorties that include use of the robots and surface mobility systems. They will also perform Extravehicular Activities (EVAs) using Extravehicular Mobile Units (EMUs). Spaceflight stressors include partial gravity, surface dust, dust storms, isolation, communication delays, no resupply, and a non-circadian day-night cycle.

4.2.1. Geological EVA Expedition Preparation (Nominal)

Table 6 presents a nominal scenario. Two crewmembers have descended to the Mars' surface. Seven sols have passed to allow them to adjust to the partial gravity. The Commander and FE-2, the planetary geologist, have been assigned an extravehicular activity (EVA) to collect scientific data about Marsquakes to gain a better understanding of Mars' interior structure. Earth-based planetary geologists have located a large bedding surface of interest based on seismology data collected by the InSight rover and satellite images of Mars' terrain that will be the focus of the data collection. The goal is to identify the most geologically advantageous locations and route for the crew to take on the EVA to optimize sampling. Route constraints include access to significant sunlight, which provides minimal temperature variations and potentially the only power source; continuous time-delayed communication with Earth via relay satellites for mission support communications and mild grading and surface debris for walking or roving mobility.

Reminder: This narrative is a descriptive story meant to provide a vivid picture to the reader. Specific technologies are only representative and are not intended to indicate requirements.

Table 6
EVA Preparation.
Each row in the following table represents a point in time with time flowing downward in the table. The first column provides the scenario. In this scenario, all operations are nominal. The second column identifies the CPSS enabling capability (see Section 3.4.3.2) and a representative technological function. TS = Task Support; WMS = Work Management Support; CS = Collaboration Support; DS = Decision Support. The rightmost column provides the exploration technology for which the CPSS has levied a requirement. These disciplines were described in Section 3.4.4, Table 4.
Two crewmembers have been assigned to study Mars geology.

Scenario Inspired by [30,31]	**CPSS Enabling Capabilities and *Technology (italics)* demonstrated**	**Exploration Technology for which the CPSS levies requirements**
Several days prior to the EVA, the Commander charges the Unmanned Aerial Vehicle (UAV) batteries to full capacity using the habitat power supply followed by pre-deployment health checks on the UAV.	WMS (*Computerized Procedures*)	Power & Energy Storage Exploration Destination Systems
One day prior to the planned EVA, the Commander, who is a pilot, deploys the UAV to survey a large bedding surface. The bedding surface contains difficult to access areas due to cliff sections. The UAV flight path can be completely user-controlled, but in this case the Commander has set the system to attend and react to local image features in real-time, skipping areas that are either poorly exposed or homogenous and suggest multiple flyovers of important features (i.e., surface fractures). The UAV goal is to identify fractures and to collect data on fracture length, orientation, intensity, and fault intersections. The pilot's goal (along with the planetary geologist) is to monitor the flight path and terrain and to alter the UAV path if the image shows an interesting artifact other than a fracture. In this manner, the UAV and the crew are a collaborative team.	WMS (*Image Pattern Recognition*) WMS (*Measurement Capability*) CS (*Crew/System Information Exchange*)	
The images are transmitted to a data analysis capability for interpretation. Important patterns are labeled and a digital nudge alerts FE-2, the planetary geologist crewmember, to discrepancies in fault displacement patterns and prompts the consideration of alternative fault patterns. Based on a short list of prioritized goals developed by the Earth experts and FE-2, the analysis includes identification of areas with sedimentation of interest for the EVA crew to sample.	DS (*Analytics*) DS (*Integrated Visualizations*) CS (*Crew/System Information Exchange*)	
The Surface Navigation System generates and visualizes an optimized route, with planned dig sites for the EVA. The system also provides the constraints that led to that route and four dig sites, the tools needed at each site and potential procedures for the digs. The system estimates that to meet all the goals set by the geologists will result in a 9-hour EVA. The geologist removes the lowest priority goal, and the system recalculates a new set of parameters for a 5.5-hour EVA. The geologist approves the route.	WMS (*Integrated Procedures, Schedule & Inventory*)	Autonomous Systems Navigation
The team retrieves the recommended computerized procedure for the first dig site. The procedural constraints indicate that it was written for an EVA team of three crewmembers. She chooses to have the system automatically edit the steps to accommodate two crew. The team and the computerized procedure generator edit the remaining procedures until they are satisfied, and the system detects no errors or infractions.	WMS (*Computerized Procedures*) WMS (*Procedure Writing Capability*)	

4.2.2. Geological EVA: Surface Navigation System (Unanticipated)

Table 7 presents an unanticipated scenario on the planetary surface. Two crew members (FE-2, the planetary geologist and the Commander) have been traversing the Mars surface on a geological expedition that has taken them 8 miles from the Mars Habitat. They have been avoiding many rocks and boulders in the battery powered rover for the past 7.5 hours (they are 1.5-hours over the planned EVA timeline). They have accomplished all the planned tasks and are ready to return.

Reminder: This narrative is a descriptive story meant to provide a vivid picture to the reader. Specific technologies are only representative and are not intended to indicate requirements.

Table 7
Geological EVA.
Each row in the following table represents a point in time with time flowing downward in the table. The first column provides the scenario. In this scenario, an unanticipated event has occurred (i.e., tasks took longer than planned to accomplish). The second column identifies the CPSS enabling capability (see Section 3.4.3.2) and a representative technological function. TS = Task Support; WMS = Work Management Support; CS = Collaboration Support; DS = Decision Support. The rightmost column provides the exploration technology for which the CPSS has levied a requirement. These disciplines were described in Section 3.4.4, Table 4.
Two crewmembers are in a rover on the Mars surface and are ready to return after 7.5 hours.

Scenario Inspired by [32,33]	CPSS Enabling Capabilities and *Technology (italics)* demonstrated	Exploration Technology for which the CPSS levies requirements
The Commander is driving the rover. FE-2 activates the Surface Navigation System (SNS) and requests a return route. The SNS pulls data on identified obstacles, surface topography, rover battery charge, crew oxygen levels and crew vitals.	DS (*Knowledge Capture & Management*) DS (*Knowledge of Context*) TS (*Proactive Aiding*)	Navigation
These data are shown to the EVA crew and is also sent to the Mars Habitat. There are 40 and 47 minutes of breathable air left in their respective EMUs. The rover battery is not fully charged due to dust obscuration.	TS (*Smart Information Presentation*)	Communication
Because of these two time-limitations, the SNS presents two recommended routes as well as limitations in retracing their route to the habitat. Although all obstacles were logged and are now known on the outgoing route, this route would take longer than allowable. Of the two recommended routes, there is a 2-mile section where the topography is known (through satellite imagery or a pre-deployed unmanned aerial vehicle), but the obstacles have not been mapped. This route will take them 20 minutes to traverse at a specified speed to reach the Habitat. The final recommended route has a 0.75-mile unknown section of obstacles but will take approximately 35 minutes to traverse at a specified speed.	DS (*Knowledge of Context*) DS (*Routing*)	
Because they are fatigued, the crew choose the route with the least uncertainty, although it will take them longer to reach the habitat.	CS (*Crew-System Collaboration*)	

4.2.3. Robot Rescue (Unanticipated) – Scenario inspired by [32,33]

Table 8 presents one solution to the Section 4.2.2 anomaly. All crew are fatigued, and sleep deprived, attempting to run as many experiments as possible during their 30-day stay on the Mars surface. FE-2 and the Commander are out on a final EVA, collecting rock samples about a half mile from the surface habitat. They have taken a robot for drilling in a rubble filled area.

4.3. Return to Earth

On the return to Earth, spaceflight stressors are like those experienced *en route* to Mars. The spaceflight stressors effects on crew performance may be exacerbated on the transit back to Earth. They will have been away from family and friends and confined to a small space with the same few crew members for months. In addition, needed supplies and food choices will be limited.

The Numa habitat and the Fasti service module are two weeks into the return trip to Earth. On the final EVA, Bill suffered a

Table 8
Robot Rescue.
Each row in the following table represents a point in time with time flowing downward in the table. The first column provides the scenario. In this scenario, an unanticipated event has occurred (i.e., one crewmember has been injured). The second column identifies the CPSS enabling capability (see Section 3.4.3.2) and a representative technological function. TS = Task Support; WMS = Work Management Support; CS = Collaboration Support; DS = Decision Support. The rightmost column provides the exploration technology for which the CPSS has levied a requirement. These disciplines were described in Section 3.4.4, Table 4.
Two crewmembers are participating in a robot-assisted EVA.

Scenario Inspired by [33-35]	CPSS Enabling Capabilities and *Technology (italics)* demonstrated	Exploration Technology for which the CPSS levies requirements
As the robot is drilling amongst the boulders and rubble, FE-2 spots an interesting hue of soil [35]. While most soil on Mars has a reddish cast, this is greenish-yellow. The Commander begins digging in that area with a shovel to acquire a sample of the soil.	CS (*Crew-Robot Information Exchange*)	Communication Robotics
The Commander abruptly falls to the ground and exclaims that his lower back feels locked, and he is unable to stand. At the time that he fell FE-2 heard a “pop” sound. FE-2 radios back to the surface habitat to speak with the crew physician. He suggests that the Commander move as little as possible.	CS (*Crew Information Exchange*)	Communication
FE-2 instructs the robot to halt drilling and configure itself for rescue operations.		Exploration Destination Systems Robotics
The robot shape-shifts, concealing the drill and unveiling a gurney. In the meantime, a procedure for safely moving and securing the Commander onto the gurney is made accessible to FE-2.	WMS (*Computerized Procedures*)	Exploration Destination Systems Robotics
FE-2 and the Commander (being transported by the robot) reach the surface habitat where appropriate medical diagnosis and treatment may be applied.	DS (*Troubleshooting*) DS (*Integrated Visualizations*) DS (*Knowledge Capture & Management*)	Human Health, Life Support and Habitation Systems

bone fracture in the lumbar region of his spine and is on strong pain medication. The other three crewmembers have a busy schedule. Bill has given the command position over to Dimitri for the return trip. When Dimitri has completed a routine course correction, he will perform a routine check on the power management and distribution sub-system. Helen, FE-2, is processing a return geological sample. Christian, FE-3 (the physician), is taking care of his patient, has just completed a load of laundry for the crew [33] and will soon be repairing the waste hygiene component (again).

The health of several crewmembers has been an issue. Dimitri has had three urinary tract infections that put him out of commission until the antibiotics take their course. FE-2's vision has continued to decline. While on the Mar's surface, her Spaceflight Associated Neuro-Ocular Syndrome (SANS) symptoms seemed to level off, but now that they are again in microgravity, her fundoscopy images show numerous cotton wool spots and severe optic disc cupping.

4.3.1. Radiation Event (Unanticipated)

Reminder: This narrative is a descriptive story meant to provide a vivid picture to the reader. Specific technologies are only representative and are not intended to indicate requirements (Table 9).

5. Conclusions

This Concept of Operations contains a description of how crew and ground habitat performance support systems could assist astronauts, cosmonauts, or taikonauts in resolving anomalies during the phases of a crewed mission to deep space destinations, such as Mars. Scenarios were used to promote a shared understanding of processes, practices and technological goals needed for safe and productive manned missions beyond low-earth orbit (LEO). This operational perspective should stimulate requirements development related to crew use of the systems.

Currently, crew receive performance support in the form of procedures, training, scheduling, inventory tracking and user interfaces. The integrated CPSS proposes to advance these capabilities and to integrate them with additional capabilities needed to support an autonomous crew. For example, on-board systems could monitor sensor output to flag anomalous readings and provide appropriate response options to the crew. The first order of change is that operations must shift from Earth-based to space-based since real-time control is not possible. Today there are sensors and trending, with many engineers and scientists analyzing these data for diagnosis and problem-solving. It will not be possible to train four astronauts the deep system knowledge needed to respond to the range of safety- and time-critical anomalies that

Table 9
Radiation Event.
Each row in the following table represents a point in time with time flowing downward in the table. The first column provides the scenario. In this scenario, an unanticipated event has occurred (i.e., solar flare alert). The second column identifies the CPSS enabling capability (see Section 3.4.3.2) and a representative technological function. TS = Task Support; WMS = Work Management Support; CS = Collaboration Support; DS = Decision Support. The rightmost column provides the exploration technology for which the CPSS has levied a requirement. These disciplines were described in Section 3.4.4, Table 4.
NASA's Solar Dynamics Observatory (SDO) has sent an alert for potential solar activity.

Scenario Inspired by [30,36]	CPSS Enabling Capabilities and *Technology (italics)* demonstrated	Exploration Technology for which the CPSS levies requirements
Earth-Support sends the Numa crew a solar weather warning.	CS (*Earth-Support-Crew Collaboration*)	Communication
The CPSS receives the message and automatically displays the individual crew's procedure for Solar Energetic Particle (SEP) events within the transit habitat. This triggers (with concurrence from the Commander) the self-scheduling tool to clear the schedule until further notice. The scheduling capability alerts the Commander to a high priority item on the schedule for the following day. The Commander acknowledges the alert and selects it to be a Watch Item, but will have to deal with the issue after the solar event.	TS (*Proactive Aiding*) WMS (*Computerized Procedures*) WMS (*Integrated Procedures & Schedule*) DS (*Watch Item*)	
Making use of the available mass, the crew begin to build a temporary shelter. FE-2 repositions storage units, while the Commander completes the course correction and turns off sensitive equipment. FE-3 stows the tools they have been using and begin to reposition food and water supplies. FE-3 also retrieves biomedical tools. FE-2 then ensures the barricade contains all the necessary supplies they will need while they shelter.		Propulsion
As the crew work through their respective procedures, they indicate completion of each step. The computerized procedure detects a missed step and alerts the Commander of this omission. The Commander completes the overlooked step. They position Bill in the radiation-protection area and the rest float into the bay. The Commander checks the security of the barrier.	WMS (*Computerized Procedures*)	
While in the bay, the CPSS suggests that the physician and the backup CMO, FE-2, review information about diagnosing and treating radiation exposure.	TS (*Smart Tutoring*)	
Thirty-six hours later, equipment on-board and on the space craft exterior detect and record high levels of radiation that lasts for 47 min. The CPSS alerts the crew when it is safe to exit the bay.	WMS (*Smart Alerts*)	Sensors & Instruments
The crew exit the protective bay and begin the post-SEP procedures. They realize that the flare has disrupted power to multiple displays and instruments including the communication capability with Earth-Support. Fortunately, the CDSS itself has not been affected.	WMS (*Computerized Procedures*)	
The Commander requests the schematics for the affected systems. The knowledge database also provides drawings of how the different systems interface with one another. Using his expertise, he surmises that the solar particles may have disrupted communication electronics.	DS (*Knowledge Management*) TS (*Proactive Aiding*) CS (*Problem-Solving Prompts*)	
The Commander pulls up the scheduling tool, requests an updated schedule including a high-priority EVA to repair the communication electronics. The scheduling tool provides a strawman schedule that also includes the Watch Item from before the solar event. The Commander is reminded of this high priority item that is highlighted on the schedule.	WMS (*Self-Scheduling*) DS (*Watch Item*)	

may occur. The CPSS could be an aiding system, using contemporary data analysis and modeling approaches. The CPSS capabilities will change the crew's job, their interactions with habitat systems, and Earth-Support's deeply supportive role. The introduction of these changes will require an intentional cultural evolution and strong leadership. Iterative research will be needed to determine how a CPSS can effectively support crew anomaly response. The CPSS provides operational support to the crew for the execution of activities as well as maintains Earth-Support situation awareness. This ConOps is intended to spark iterative responses from the users, system designers, researchers, programs, and space agencies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Bettina L. Beard: Methodology, Project administration, Supervision, Visualization, Writing – original draft.

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